Evaluating Transaction Failure Risk for Solana Signers Without Available History

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Abstract-Historical signer outcomes can provide strong aggregate predictions of blockchain transaction failure while offering little information when signer history is unavailable. We study this problem on Solana using declared account roles and delayed historical outcomes. Four chronological evaluation windows contain approximately 1.59 million sampled transaction records across twelve dates. A role-separated history representation summarizes failures involving non-signer writable accounts and augments a gradient-boosted baseline containing transaction structure, signer history and read-only account and template context. Cold-history average precision improves in all four windows, although the confidence interval includes zero in one window. Account-identity controls yield mixed results, motivating a narrower first-appearance evaluation. On a reserved test containing 17,904 first signer appearances, writable history increases average precision from 0.1688 to 0.2004 after both models receive the same delayed account-identity score. The paired hourly bootstrap interval for the difference is [0.0168, 0.0501]. Ranking improvement does not imply better operational risk estimates: inherited validation thresholds produce high false-positive rates, and calibration deteriorates in this comparison. The contribution is an empirical evaluation of missing-history reliability prediction and a reproducible account-context benchmark, rather than a new learning algorithm or demonstrated transaction-failure prevention system.

Index Terms-Solana, transaction reliability, machine learning, cold start, delayed feedback, account context.

I. INTRODUCTION

Transaction reliability is an operational concern for applications interacting with a blockchain. A submitted operation may fail during execution even after it reaches a block. On Solana, a large empirical study has characterized failure initiators, programs, error categories and temporal patterns [1]. These descriptive findings motivate predictive evaluation, but a model's aggregate performance does not establish its usefulness for a signer with no available outcome history.

A transaction can involve accounts already observed in other transactions even when its signer is unfamiliar to an observer. We ask whether failures involving these accounts supply transferable execution-risk signal. The hypothesis concerns observed participation, not ownership: two signers touching the same account need not belong to the same operator. Nor does a shared account's historical failure rate identify the cause of a new failure.

The task is retrospective execution-risk screening conditional on eventual block inclusion. The archive supplies inclusion checkpoints and outcomes but not actual submission times or unlanded attempts. We restrict predictive inputs to declared account information and earlier outcomes whose assumed availability time has elapsed. This prevents current execution receipts from supplying the answer, while preserving the limitation that the checkpoint is reconstructed after inclusion.

The study contributes a concrete benchmark for transactions with unavailable signer history, an ablation of writable-account history against structure and read-only context, and a reserved first-appearance comparison with learned account-identity scores. It also documents the distinction between improved failure ranking and poor calibration. History aggregation and gradient boosting are established techniques; algorithmic novelty is not claimed.

II. RELATED WORK

Zheng et al. analyze failure initiators, programs, errors and temporal patterns on Solana [1]. Alizadeh and Khabbazian study its transfer-network structure and failed transfers [4]. These studies motivate reliability analysis, while our comparison tests execution-outcome ranking for signers without available history.

Relational risk transfer is established. BRIGHT exploits historical entity context for real-time fraud learning [2]. Our task concerns execution reliability and tabular participation summaries; neither relational learning nor restrictions on future information are new contributions.

Chaudhary et al. evaluate Ethereum multisignature anomaly triage after execution, using receipt-derived features [5]. Our inputs exclude current execution receipts and our target is the observed Solana success/failure outcome. The contribution is a narrower evaluation setting and empirical result, rather than the first blockchain classifier or a new boosting method.

TABLE I. CLOSEST WORK AND TASK DISTINCTIONS

Work	Established focus	Present distinction
Zheng et al. [1]	Solana failure characterization	Missing-history predictive evaluation
Lu et al. [2]	Relational fraud learning	Execution outcome rather than fraud
Alizadeh et al. [4]	Transfer-network analysis	Declared-role history ablation
Chaudhary et al. [5]	Post-execution anomaly triage	Exclude current receipt features

III. PROBLEM DEFINITION AND DATA

We use public transaction records from SolArchive [3], under CC BY 4.0, pinned to Hugging Face revision 6588cad2832e62be03972ac80d69c6af11d631c8. Each selected date contributes the first row group of one Parquet shard. The first two windows use the lexicographically first shard; later windows use alternate shards selected by a fixed date and seed hash based on file counts. Selection does not depend on transaction outcomes. This is neither complete chain coverage nor a verified uniform random sample.

Selected fields are transaction signature, declared accounts with signer and writable flags, block timestamp, block slot and status. Block slot and signature order tied checkpoints; neither is a model feature. Status maps Success to zero and Fail to one. Identical signatures with identical selected payloads count once; conflicting duplicates or date mismatches abort processing. Fifteen identical repeated rows occur in the second date of the initial window. The twelve extracted groups contain 1,588,408 raw rows before deduplication and model partition exclusions.

We call the first declared signer the primary signer. A **cold-history transaction** has no matured outcome for that address as primary signer in the sampled replay. It may be a repeat transaction while earlier outcomes remain pending. A **first appearance** is the address's first occurrence as primary signer in that replay, regardless of label maturity. Neither definition establishes that an address is new to Solana or that its operator is new.

The archive does not contain complete instruction payloads in the selected table. Read-only account templates are sets of declared read-only accounts, not verified lists of invoked programs. Historical account resolution before execution is assumed. The later prototype constructs declared roles directly for plain SOL transfers and reads parsed account roles from public RPC blocks; this does not reconstruct historical submission conditions.

IV. SYSTEM ARCHITECTURE AND MODELS

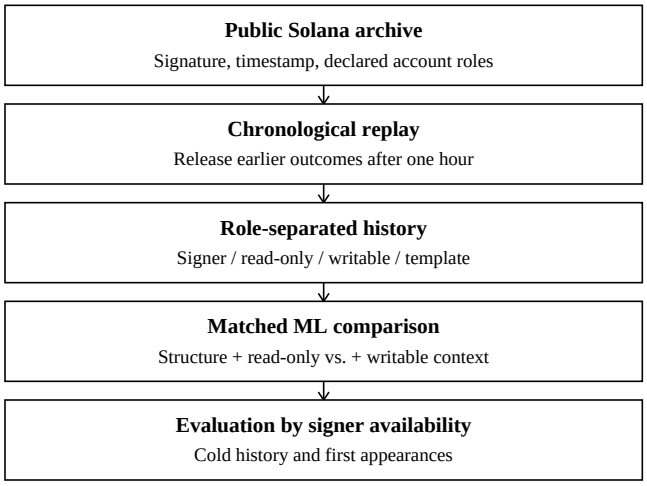

Fig. 1. Implemented replay and matched account-context evaluation. Current execution metadata never enters the feature extractor.

At checkpoint time t, an earlier transaction contributes its label only if its block checkpoint plus 3,600 seconds is no later than t. Features are computed before scheduling the current outcome. Histories update during validation and testing using earlier matured outcomes, making the evaluation an online-feedback replay with a fixed fitted boosting model.

Account participation is separated into primary signer, non-signer writable accounts, non-signer read-only accounts and exact read-only templates. Ten common infrastructure addresses are excluded from contextual histories and retained as structural presence indicators. For each role and account key, a smoothed historical rate is (failures + 20 x global_rate) / (count + 20), where global_rate = (available_failures + 1) / (available_count + 2). Multi-account roles are summarized through observed fractions, log-count means and maxima, and rate means, maxima, minima and standard deviations.

Structural features contain account and role counts, duplicate-account counts and infrastructure indicators. Current labels, logs, errors, actual consumed compute, balances, paid fees and block position are excluded. Time controls history availability and partitions, but is not a numeric model input.

Seven original models comprise signer-rate, read-only-rate and template-rate scorers; structure-only boosting; structure plus signer history; structure plus signer and read-only/template history; and all preceding features plus writable history. Histogram gradient boosting uses 150 iterations, at most 15 leaves, minimum leaf size 40, L2 regularization 5 and no early stopping. Three seed predictions are averaged; deterministic fitting means these seeds are not independent replications.

The identity control is an SGD logistic learner with role-tagged account identities and read-only templates hashed into 1,048,576 nonnegative dimensions. Binary

tokens are L2 normalized. It uses regularization 0.0001, seed 42 and one chronological pass. Predictions precede each batch's learning update. Labels become eligible after one hour, with ten-minute update batches adding up to ten minutes of further delay. The resulting prequential score is supplied to both history-based boosting models. Thus the stronger baseline receives information from all declared account identities, including writable identities; the tested addition is the explicit writable-history summaries. The identity architecture and regularization are fixed rather than exhaustively tuned.

V. MATHEMATICAL FORMULATION

Let transaction i have checkpoint time t_i, outcome y_i in {0,1}, and declared account-role sets. The available history at t contains only transactions with t_i + 3,600 <= t. For role r and key a, let n_r(a,t) be its available participation count and f_r(a,t) its failure count. The smoothed estimate is:

$$q_r(a,t) = [f_r(a,t) + 20\, b(t)] / [n_r(a,t) + 20] \quad (1)$$

Here b(t) = [F(t) + 1] / [N(t) + 2] is the smoothed global failure rate. The candidate augments baseline feature vector x with summaries of q over non-signer writable accounts. Both models use the same matured outcomes and, in the identity-controlled comparison, the same prequential identity score.

The principal ranking contrast is Delta AP = AP(y, s_context) - AP(y, s_baseline), evaluated on the identical subgroup. Average precision aggregates precision at positive-label recall increments. A gain is a ranking result; it does not guarantee calibration or better recall at a shared false-positive rate. Brier loss is the mean squared difference between predicted probability and y; lower values are better.

VI. EXPERIMENTAL METHODOLOGY

Each window uses one training date, one validation date and one test date. The first six training hours build history. Training and validation rows are excluded when their labels would not be available before the next date boundary. Test labels evaluate predictions and update later histories only after the stipulated delay. Models are fitted separately in each window using unchanged parameters.

TABLE II. CHRONOLOGICAL PARTITION SIZES

Test date	Train	Validation	Test
11-03	92,782	125,864	131,759
11-10	92,135	126,214	133,071
11-24	93,812	127,050	131,638
12-01	100,517	126,840	131,907

All dates are in 2025. Each training and validation day immediately precedes its test day. Full windows are November 1-3, November 8-10, November 22-24, and November 29-December 1.

The original primary endpoint is the cold-history difference in positive-class average precision between the read-only-context baseline and writable-context model. The first confirmation retains the initial settings. The extension follows after the first two results. Identity controls and first-appearance analyses are exploratory on these three windows.

Before inspecting the fourth window's transaction outcomes, a refined endpoint is fixed: first-appearance average precision for writable versus read-only history when both models receive the identity score. The local success condition requires a positive difference and a paired-hour interval entirely above zero. This is a reserved historical evaluation, not public preregistration or a live prospective trial. The change in hypothesis follows earlier diagnostics and is explicitly disclosed.

We report average precision, AUROC, Brier loss and operating metrics. Thresholds are chosen on global validation data under a 5% false-positive budget and then held fixed. Differences in average precision use 500 paired bootstrap draws of hourly blocks within the test day. Signer-address resampling supplies an additional dependence diagnostic. These intervals condition on fitted predictions and sampled days; they exclude retraining uncertainty and independent operator identification.

VII. RESULTS AND ANALYSIS

Signer history alone gives overall average precision between approximately 0.80 and 0.83, but cold-history scores of 0.062-0.073 are close to subgroup failure prevalence. This is partly structural: absent signer outcomes, the scorer mostly returns the changing global prior. Aggregate performance therefore cannot substitute for an evaluation of missing-history transactions.

TABLE III. ORIGINAL COLD-HISTORY RANKING

Date	Cold rows	Base AP	Context AP
11-03	19,621	0.3193	0.3787
11-10	20,636	0.2249	0.2514
11-24	21,028	0.2052	0.2231
12-01	19,577	0.1734	0.2185

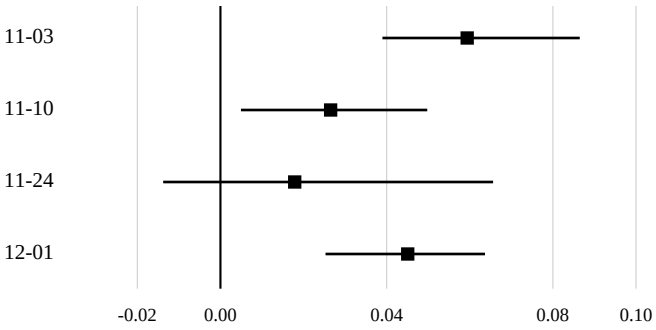

Fig. 2. Original cold-history AP difference and paired hourly 95% intervals. November 24 includes zero; this is not the refined identity-controlled endpoint.

All point differences are positive, but the extension interval crosses zero. Adding identity scores produces cold-history gains of approximately 0.0718, 0.0041, 0.0353 and 0.0391 across the four windows. The second interval crosses zero. These results prevent a claim of universal advantage beyond learned account identities.

On the reserved first-appearance endpoint, 17,904 transactions have failure prevalence 5.88%. The identity-controlled read-only baseline reaches average precision 0.1688; writable context reaches 0.2004. Their difference is 0.0316 with hourly 95% interval [0.0168, 0.0501], satisfying the locally specified ranking criterion. AUROC rises from 0.7288 to 0.7545.

Operational and calibration results are less favorable. At inherited global-validation thresholds, first-appearance false-positive rates are 22.66% for the baseline and 28.65% for writable context. Recall rises from 50.90% to 64.48%, but this is not a matched-false-positive comparison. Brier loss worsens from 0.0967 to 0.1218. The stronger ranking model therefore cannot be described as an overall better probabilistic predictor or alert system. These results also show why validation-only threshold selection does not guarantee a test-period budget.

TABLE IV. RESERVED FIRST-APPEARANCE TEST

Metric	Baseline + ID	Context + ID
Average precision	0.1688	0.2004
AUROC	0.7288	0.7545
Brier loss	0.0967	0.1218
False-positive rate	22.66%	28.65%
Recall	50.90%	64.48%

VIII. IMPLEMENTATION AND OPERATIONAL FOLLOW-UP

A fifth chronological window uses December 3 for training, December 4 for validation and December 5, 2025 for testing. Its settings were fixed before those outcomes were inspected. On 19,515 first appearances, the identity-controlled baseline and writable-context models achieve average precision 0.2207 and 0.2393, respectively; the difference is 0.0186 with paired-hour interval [0.0095, 0.0288]. This supports the ranking result without establishing operational utility.

A subsequent exploratory policy audit evaluates 102,443 cold-history transactions, including 6,410 failures, across all five test dates. Exported non-identity HGB models use isotonic calibration fitted on early cold validation data and thresholds chosen on later cold validation data. On December 5 the deployed context threshold warns on 357 failures and 649 successes. Fixed cost-based thresholds are also evaluated: for assumed failed-attempt fee F and cost C of blocking a success, the theoretical probability threshold is C/(F+C). Calibration is imperfect, so the rule need not deliver its assumed utility. With C/F=5, pooled context warnings cover 107 failures and 60 successes and miss 6,303 failures; hypothetical net utility TP-5FP is -193 fee units. All three evaluated positive cost ratios (1, 5 and 10) yield negative pooled utility. These are assumed costs, not paid fees or measured savings. The audit is retrospective and must not be promoted to an independent confirmation.

The integrated FastAPI prototype constructs fresh features with the same participation-history extractor, using mature public RPC block samples. At verification it collected 302 devnet transactions in 16 sampled blocks and 8,871 mainnet transactions in eight sampled blocks, spanning roughly eight hours per network. Earlier outcomes must be at least one hour old, a snapshot must be no older than one hour, and minimum history requirements are enforced. Actual HGB inference runs on the resulting feature row. Because source, sampling and operating conditions differ from training, outputs are explicitly experimental model scores: no validated live probabilities or automatic ML blocking decisions are supplied. Central training-range diagnostics do not establish distributional equivalence.

Current balance and fee checks and an unsigned simulation accompany each candidate. Signature verification is disabled; this observation is distinguished from a signed submission's normal preflight. A local SQLite study store records the candidate message hash, feature/history/model hashes and prediction before an outcome can be attached. A confirmed receipt is accepted only for the exact checked message and a later slot and timestamp. Missing receipts remain unknown outcomes. Observed paid fees and elapsed observation intervals are recorded without relabelling them as savings.

A controlled local-validator experiment proves the fee mechanism: an execution failure deliberately submitted with

preflight disabled pays 5,000 lamports, while a matched checked-and-stopped attempt pays zero. Normal preflight also rejects that error at zero cost. A separate local successful transfer verifies the decision-to-matching-receipt workflow. These functional checks establish neither superiority over standard preflight nor ML-driven fee or time savings. A prospective study of candidates passing standard preflight, followed by an authorised policy comparison, is still needed for those claims.

IX. DISCUSSION AND LIMITATIONS

The findings support a bounded interpretation: explicit account-participation histories can improve ranking for sampled signers without outcomes, including a reserved first-appearance comparison after an identity-score control. They do not establish that writable accounts cause failure or that the gain extends to all applications and market regimes. Some failed operations may be protective rejections, so reducing failures is not automatically a beneficial intervention.

Sampling is a central limitation. One group per date may reflect partitioning biases, and a sample-cold address may have extensive unsampled history. The archive's dated inventory is discontinuous. All windows come from the same provider and a limited period, leaving independent-source generalization unresolved. Range and derived-file hashes support extraction reproducibility, but original complete-shard hashes have not been verified.

The information boundary is also limited. Inclusion time is not submission time and unlanded attempts are absent. The later live implementation covers plain SOL transfer roles, not arbitrary historical account resolution. The identity control uses batching while exact participation histories do not, introducing a feedback mismatch. Additional validation-tuned identity models and matched update schedules could alter the comparison.

Hourly resampling captures some within-day temporal dependence but not uncertainty across independently sampled regimes. Address clustering does not identify operators. The refined endpoint follows earlier exploratory findings; its reserved confirmation supplies additional evidence without making the earlier analyses confirmatory. No multiplicity-adjusted claim is made across all diagnostics.

For practical use, cold-specific calibration and independently tested operating thresholds are required. Comparison against live transaction simulation additionally requires actual submission, simulation and execution observations. The current archive cannot demonstrate superiority to simulation, fee savings or prevented failures.

X. CONCLUSION AND FUTURE WORK

A high aggregate score from signer history conceals weak information for transactions without available signer outcomes. On sampled Solana records, shared writable-account participation supplies additional ranking signal beyond structure and read-only context. A reserved first-appearance test retains a positive gain after a learned identity-score control. Calibration and false-alarm performance constrain the result's operational interpretation. The study offers a reproducible empirical benchmark and evidence for account-context transfer under missing signer history, with broader validation needed before deployment claims.

REPRODUCIBILITY

Data from SolArchive.org [3], CC BY 4.0, pinned revision 6588cad2832e62be03972ac80d69c6af11d631c8. Local code, protocols, dependency versions and reports accompany this manuscript in the project repository. Raw transaction files remain local. An immutable public artifact URL remains to be supplied for submission. Source hashes and code hashes are recorded per experiment.

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